

Probabilistic Classifiers

Procheta Sen



Probabilistic vs “Ordinary” Classifier

- ▶ **“Ordinary” classifier** is a function f that assigns to an input object $\mathbf{X}$ a predicted class c from a fixed set of classes $\{c_1, c_2, \dots, c_k\}$, i.e.

$$c = f(\mathbf{X}).$$

- ▶ **Probabilistic classifier** is a **conditional distribution** $P(C | \mathbf{X})$. For an input object $\mathbf{X}$ it gives probabilities $p_1, p_2, \dots, p_k$, where

$$p_i = P(c_i | \mathbf{X})$$

and

$$p_1 + p_2 + \dots + p_k = 1.$$

Two types of Models: Discriminative vs Generative

Discriminative

- ▶ Assume that the conditional distribution $P(C | X)$ (i.e. the probabilistic classifier) has specific form $P_{\theta}(C | X)$ depending on some parameters

$$\theta = (\theta_1, \dots, \theta_k)$$

- ▶ Use training data set to **find / learn** parameters $\theta_1, \dots, \theta_k$ such that the resulting distribution is “best possible” among all distributions of the assumed form

Generative

- ▶ Assume that data come from specific distribution $P_{\theta}(X, C)$ depending on some parameters

$$\theta = (\theta_1, \dots, \theta_k)$$

- ▶ Use training data set to **find / learn** parameters $\theta_1, \dots, \theta_k$ such that the resulting distribution is “best possible” among all distributions of the assumed form
- ▶ Use $P_{\theta}(X, C)$ to classify new objects

Generative Models

Data generating distribution

- ▶ **Assumption:** data come from some unknown probability distribution P over object-class pairs $(\bar{X}, c) \in \mathcal{X} \times \mathcal{C}$, i.e. the classification problem is characterised by P .
- ▶ **Suppose we know P :** for any pair $(\bar{X}, c) \in \mathcal{X} \times \mathcal{C}$ we can compute $P(\bar{X}, c)$, i.e., the probability of the event that an object with feature vector $\bar{X}$ belongs to class c .

Then we can compute the **Bayes optimal classifier**

$$f^*(\bar{X}) = \arg \max_{c \in \mathcal{C}} P(\bar{X}, c)$$

The Bayes optimal classifier

$$f^*(\bar{X}) = \arg \max_{c \in \mathcal{C}} P(\bar{X}, c)$$

The Bayes optimal classifier is best possible, in the sense that if g is another classifier, then for any $\bar{X} \in \mathcal{X}$ the probability that f^* errs on $\bar{X}$ is smaller than the probability that g errs on $\bar{X}$

In practice...

- ▶ We do not know the data generating distribution P
- ▶ Instead, using training data, we can try to learn a distribution $\hat{P}$, which we hope to be very similar to P
- ▶ And then use $\hat{P}$ for classification

One of the ways to construct $\hat{P}$

- ▶ Assume that the distribution P belongs (or is “close” to) some family of parametric distributions (e.g. Normal distribution) (**the model assumption**)
- ▶ **Estimate parameters** which give specific distribution in the family that is most likely to be the generating distribution for the training data
- ▶ Key assumption: the training data is drawn from P independently, i.e. the training instances $(\bar{X}, c)$ are **independently and identically distributed** (the i.i.d. assumption)

Normal distribution $\mathcal{N}(\mu, \sigma)$ is **parametric**:
the two parameters are mean (μ) and standard deviation (σ)

Estimation of parameters

A common approach to estimate parameters is

Maximum likelihood estimation method

Choose the parameters that **maximise**
the **probability** (i.e. **likelihood**) of the observed (i.e.
training) data

Maximum likelihood estimation: single parameter

Example 1

- ▶ Suppose we want to model a **biased coin** (or a **class** in binary classification problem)
- ▶ and observe data $THHH$, where H – heads, T – tails.
- ▶ Assume further:
 - ▶ All the flips came from the same coin and each flip was independent (the i.i.d. **assumption**)
 - ▶ The coin has a fixed probability β of coming up heads and the probability $1 - \beta$ of coming up tails, i.e. we assume that the data generation distribution is from the family of **Bernoulli distributions parameterised** by the probability of heads $\beta \in [0, 1]$ (the **model assumption**)

Maximum likelihood estimation: single parameter

Example 1

- ▶ Observed data: $THHH$
- ▶ For the heads probability β the probability of observed data:

$$\begin{aligned}P_{\beta}(THHH) &= P_{\beta}(T) \cdot P_{\beta}(H) \cdot P_{\beta}(H) \cdot P_{\beta}(H) \\&= (1 - \beta)\beta\beta\beta = (1 - \beta)\beta^3 = \beta^3 - \beta^4\end{aligned}$$

Maximum likelihood estimation: single parameter

Example 1

- ▶ To find β that maximises
 $P_{\beta}(THHH) = \beta^3 - \beta^4$

$$\frac{\partial}{\partial \beta}[\beta^3 - \beta^4] = 3\beta^2 - 4\beta^3$$

- ▶ compute the derivative of $\beta^3 - \beta^4$

$$3\beta^2 - 4\beta^3 = 0 \iff 3\beta^2 = 4\beta^3$$

- ▶ set it equal to 0

$$\beta = \frac{3}{4}$$

– the maximum likelihood estimate of β

- ▶ and solve for β

Maximum likelihood estimation: single parameter

Example 2 (generalisation)

- ▶ Assume that the observed data consists of h heads and t tails.
- ▶ Then the probability of the observed data (under a Bernoulli model with head probability β) is

$$P_{\beta}(\text{data}) = \beta^h (1 - \beta)^t.$$

- ▶ Instead of maximising the probability directly, it is often more convenient to maximise its logarithm (the *log-likelihood*):

$$\ell(\beta) = \log(\beta^h (1 - \beta)^t) = h \log \beta + t \log(1 - \beta).$$

Exercise Compute the maximum-likelihood estimate of β (i.e. find the $\beta \in (0, 1)$ that maximises

Maximum likelihood estimation: multiple parameters

Example 3

- ▶ Suppose we want to model a K -sided die (or a **class** in **multiclass** classification problem).
- ▶ We can model this with parameters $\beta_1, \beta_2, \dots, \beta_K$, where β_i is the probability of having the i -th side of the die. Since β 's are probabilities, we should also assume:
 - ▶ $\beta_i \geq 0$ for every $i = 1, \dots, K$, and
 - ▶ $\beta_1 + \beta_2 + \dots + \beta_K = 1$.

The model assumption is that the data generating distribution is from the parametric family of

**generalised Bernoulli
distribution**

Maximum likelihood estimation: multiple parameters

Example 3

- ▶ Suppose the observed data consists of 1 rolls of 1, 2 rolls of 2, and so on.
- ▶ Then the probability of this data is

$$1^1 \cdot 2^2 \cdots K^K.$$

- ▶ The log probability (i.e. log likelihood) is displayed below:

$$\sum_{i=1}^K x_i \log \beta_i$$

- ▶ Thus to find the maximum likelihood estimate of β 's we need to find β 's that maximise the log likelihood subject to the constraint $\beta_1 + \beta_2 + \cdots + \beta_K = 1$.

Maximum likelihood estimation: multiple parameters

Example 3

$$\max_{\beta_1, \dots, \beta_K} \sum_{i=1}^K x_i \log \beta_i$$

subject to

$$\sum_{i=1}^K \beta_i - 1 = 0$$

Using the method of Lagrange multipliers we obtain the following Lagrangian

$$\mathcal{L}(\beta_1, \dots, \beta_K, \lambda) = \sum_{i=1}^K x_i \log \beta_i - \lambda \left(\sum_{i=1}^K \beta_i - 1 \right)$$

Maximum likelihood estimation: multiple parameters

Example 3

- For fixed i , we have

$$\frac{\partial \mathcal{L}(\bar{\beta}, \lambda)}{\partial \beta_i} = \frac{x_i}{\beta_i} - \lambda$$

Setting $\frac{\partial \mathcal{L}(\bar{\beta}, \lambda)}{\partial \beta_i}$ to 0 and solving with respect to β_i we get

$$\beta_i = \frac{x_i}{\lambda}$$

Maximum likelihood estimation: multiple parameters

Example 3

From the constraint $\beta_1 + \beta_2 + \cdots + \beta_K = 1$, (which is equivalent to $\frac{\partial \mathcal{L}(\bar{\beta}, \lambda)}{\partial \lambda} = 0$) we have

$$\frac{x_1 + \cdots + x_K}{\lambda} = 1$$

and hence

$$\lambda = \sum_{i=1}^K x_i.$$

Thus the maximum likelihood estimate of β 's is

$$\beta_i = \frac{x_i}{\sum_{i=1}^K x_i} \quad \text{for every } i = 1, \dots, K$$

Probabilistic classifiers

Generative

- ▶ Naive Bayes
- ▶ ...

Discriminative

- ▶ Logistic regression
- ▶ Multilayer perceptrons (neural networks)
- ▶ ...

Naive Bayes Classifier

Procheta Sen



Bayes' Rule

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where H and E are events and $P(E) \neq 0$.

Terminology

- ▶ $P(H | E)$: The probability of hypothesis H , given evidence E .
- ▶ $P(E)$: Marginal probability of the evidence E .
- ▶ $P(H)$: Prior probability of hypothesis H .
- ▶ $P(E | H)$: Likelihood of the evidence given hypothesis.
- ▶ $P(H | E)$: Posterior probability of hypothesis H .

Bayes' Rule

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where H and E are events and $P(E) \neq 0$

Bayes' Rule is useful for estimating $P(H | E)$ when it is hard to estimate $P(H | E)$ directly from the training data, but $P(E | H)$, $P(H)$, and $P(E)$ can be estimated more easily.

Bayes' Rule: derivation

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where H and E are events and $P(E) \neq 0$

By the definition of conditional probability, we have

$$P(H | E) = \frac{P(H, E)}{P(E)} \quad \text{and} \quad P(E | H) = \frac{P(H, E)}{P(H)},$$

from which we derive Bayes' Rule.

Example (single feature)

- ▶ **Meningitis** causes a **stiff neck** 50% of the time.
- ▶ Meningitis occurs $1/50000$ and stiff neck occurs $1/20$.
- ▶ Compute the probability of meningitis, given that the patient has a stiff neck.
- ▶ H = meningitis, E = stiff neck
- ▶ $P(H) = 1/50000$, $P(E) = 1/20$, $P(E | H) = 0.5$
- ▶ From Bayes' rule we have

$$P(H | E) = \frac{P(E | H) P(H)}{P(E)} = 0.0002$$

Example

- ▶ **Meningitis** causes a **stiff neck** 50% of the time.
- ▶ Meningitis occurs $1/50000$ and stiff neck occurs $1/20$.
- ▶ Compute the probability of meningitis, given that the patient has a stiff neck.
- ▶ H = meningitis, E = stiff neck
- ▶ $P(H) = 1/50000$, $P(E) = 1/20$, $P(E | H) = 0.5$
- ▶ From Bayes' rule we have

$$P(H | E) = \frac{P(E | H) P(H)}{P(E)} = 0.0002$$

- ▶ If we have 1-dimensional space (only one feature: $\bar{X} = (a)$), then we can estimate $P(H | \bar{X})$ directly from the training data set.
- ▶ It becomes more problematic if we have higher dimension. Say $\bar{X} = (a_1, a_2, \dots, a_d)$ for $d > 1$.

Example (2 features)

- ▶ Let H = engine-does-not-start
- ▶ Evidences A = weak-battery and B = no-gas
- ▶ To estimate $P(H | A, B)$ directly, we need to restrict our consideration only to those cars (objects) in the dataset that had a **weak battery** and **no gas**. Among those we need to count the cars with **non-working engine**. Such cases could be rare in our dataset making estimate of $P(H | A, B)$ unreliable or zero (in the worst case).

$$P(H | A, B) = \frac{\# \text{ weak-bat. \& no-gas \& eng.-not-working}}{\# \text{ weak-bat. \& no-gas}}$$

- ▶ Bayes' rule provides a way of expressing $P(H | A, B)$ directly in terms of $P(A, B | H)$:

$$P(H | A, B) = \frac{P(A, B | H)P(H)}{P(A, B)}$$

and estimate the latter using **naive Bayes approximation** (which is much easier to do).

Naive Bayes approximation

- ▶ Let C be a random variable representing the class of an unseen d -dimensional test object $\bar{X} = (x_1, x_2, \dots, x_d)$, where $x_1, x_2, \dots, x_d$ denote random variables of individual dimensions.
- ▶ Given a specific test object $(a_1, a_2, \dots, a_d)$ the goal is to estimate

$$P(C = c \mid \bar{X} = (a_1, a_2, \dots, a_d)) = P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)$$

- ▶ By Bayes' rule

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C=c) P(x_1=a_1, x_2=a_2, \dots, x_d=a_d \mid C)}{P(x_1=a_1, x_2=a_2, \dots, x_d=a_d)}$$

Naive Bayes approximation

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)}{P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)}$$

Does not depend on the class variable C

The class c with the largest numerator

$$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

has the largest **posterior** probability

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)$$

Naive Bayes approximation

$$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

- ▶ $P(C = c)$: Can be **estimated** as the fraction of the training data objects that belong to class c .
- ▶ How to estimate $P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$?

Naive assumption

The values of different features $x_1, x_2, \dots, x_d$ are **independent** of one another **conditional on the class**.

Independent events (2 events)

Joint probability of two events A and B

$$P(A, B) = P(A | B)P(B)$$

this holds for ANY pair of events A and B , irrespective of whether they are independent or not.

If A is independent of B , then B 's occurrence has no “consequence” on A

$$P(A | B) = P(A)$$

Thus, when A and B are **independent**, their joint probability

$$P(A, B) = P(A)P(B)$$

Independent events (> 2 events)

A finite set $\{A_1, A_2, \dots, A_n\}$ of events is **mutually independent** if for any $2 \leq k \leq n$ and for any subset of k events

$$\{A_{i_1}, A_{i_2}, \dots, A_{i_k}\} \subseteq \{A_1, A_2, \dots, A_n\}$$

we have

$$P(A_{i_1}, A_{i_2}, \dots, A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}) = \prod_{j=1}^k P(A_{i_j})$$

Naive Bayes approximation

How to estimate $P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$?

Naive assumption

The values of different features $x_1, x_2, \dots, x_d$ are **independent** of one another **conditional on the class**.

$$P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c) = \prod_{i=1}^d P(x_i = a_i \mid C = c)$$

$P(x_i = a_i \mid C = c)$: Can be estimated as the fraction of training objects in class c that have feature $x_i = a_i$.

Naive Bayes approximation

$P(x_i = a_i \mid C = c)$:

Can be estimated as the fraction of training objects in class c that have feature $x_i = a_i$

This is much easier to estimate from the training data than

$P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$

Hence, under the independence assumption, the estimation of

$P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c) = P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$

reduces to the estimations of

$P(C), P(x_1 = a_1 \mid C = c), P(x_2 = a_2 \mid C = c), \dots, P(x_d = a_d \mid C = c)$

Being naive makes life easy

Example (2 features)

- ▶ Let H = engine-does-not-start
- ▶ Evidences A = weak-battery and B = no-gas
- ▶ Direct estimation

$$P(H | A, B) = \frac{\# \text{ weak-bat. \& no-gas \& eng.-not-working}}{\# \text{ weak-bat. \& no-gas}}$$

- ▶ Using Bayes' rule + **Naive Bayes approximation**

$$P(H | A, B) = \frac{P(A, B | H)P(H)}{P(A, B)} = \frac{P(A | H)P(B | H)P(H)}{P(A, B)}$$

- ▶ $P(A | H)$ can be estimated as the fraction of cars with **weak battery** among cars with **engine not working**
- ▶ $P(B | H)$ can be estimated as the fraction of cars with **no gas** among cars with **engines not working**

Making the independence assumption makes estimates possible in practice

Bayes' rule: Proportional Form

- ▶ Assume $\overline{X} = (a_1, a_2, \dots, a_d)$ is the input test object
- ▶ We need to select/predict the class of $\overline{X}$ from the set $\{c_1, c_2, \dots, c_k\}$.

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) = \frac{P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)}{P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d)}$$

$$P(C = c \mid x_1 = a_1, x_2 = a_2, \dots, x_d = a_d) \propto P(C = c) P(x_1 = a_1, x_2 = a_2, \dots, x_d = a_d \mid C = c)$$

$$P(C \mid X) \propto P(C) P(X \mid C)$$

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X}$ = (Outlook = sunny, Temp = cool, Humidity = high, Windy = true)

$$\begin{aligned}
 P(\text{Play}=\text{yes}|\bar{X}) &\propto P(\bar{X}|\text{Play}=\text{yes})P(\text{Play}=\text{yes}) \\
 &= P(\text{sunny}|\text{yes})P(\text{cool}|\text{yes})P(\text{high}|\text{yes})P(\text{true}|\text{yes})P(\text{yes}) \\
 &= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529
 \end{aligned}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X} = (\text{Outlook} = \text{sunny}, \text{Temp} = \text{cool}, \text{Humidity} = \text{high}, \text{Windy} = \text{true})$

$$\begin{aligned}P(\text{Play=no}|\bar{X}) &\propto P(\bar{X}|\text{Play=no})P(\text{Play=no}) \\&= P(\text{sunny}|\text{no})P(\text{cool}|\text{no})P(\text{high}|\text{no})P(\text{true}|\text{no})P(\text{no}) \\&= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.020\end{aligned}$$

Example: predicting whether to play or not

Outlook			Temperature			Humidity			Windy			Play		
	Yes	No		Yes	No		Yes	No		Yes	No		Yes	No
Sunny	2	3	Hot	2	2	High	3	4	False	6	2		9	5
Overcast	4	0	Mild	4	2	Normal	6	1	True	3	3			
Rainy	3	2	Cool	3	1									
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	False	6/9	2/5		9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5	True	3/9	3/5			
Rainy	3/9	2/5	Cool	3/9	1/5									

Test instance

$\bar{X}$ = (Outlook = sunny, Temp = cool, Humidity = high, Windy = true)

Posterior comparison

$$P(\text{Play=yes}|\bar{X}) \propto P(\bar{X}|\text{Play=yes})P(\text{Play=yes}) = 0.00529$$

$$P(\text{Play=no}|\bar{X}) \propto P(\bar{X}|\text{Play=no})P(\text{Play=no}) = 0.020$$

Therefore, Play = no

Naive Bayes: classification task

In the previous setting (**choose the most probable class**):

- ▶ Given a test object $\bar{X} = (a_1, a_2, \dots, a_d)$, we wanted to predict its class C
- ▶ For this, we used the proportional form

$$P(C = c \mid X = \bar{X}) \propto P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$$

and it was enough to find $c \in \{c_1, c_2, \dots, c_k\}$ that maximises

$$P(C = c) \prod_{i=1}^d P(x_i = a_i \mid C = c)$$

Zero probabilities and Laplace smoothing

Procheta Sen



Example: predicting whether to play or not

	Outlook		Temperature		Humidity		Windy		Play	
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Sunny	2	3	Hot	2	2	High	3	4	6	2
Overcast	4	0	Mild	4	2	Normal	6	1	3	3
Rainy	3	2	Cool	3	1					
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5		
Rainy	3/9	2/5	Cool	3/9	1/5					

Test instance $\tilde{X} = (\text{Outlook} = \text{sunny}, \text{Temp} = \text{cool}, \text{Humidity} = \text{high}, \text{Windy} = \text{true})$

$$P(\text{Play=no} \mid \tilde{X}) \propto P(\tilde{X} \mid \text{Play=no})P(\text{Play=no})$$

Probability estimations

$$P(x_i = a \mid C = c) = \frac{n(a, c)}{N(c)},$$

where

$n(a, c)$ is the number of training objects in class c with $x_i = a$,

$N(c)$ is the total number of training objects in class c

Example: predicting whether to play or not (zero probabilities)

	Outlook		Temperature		Humidity		Windy		Play	
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Sunny	2	3	Hot	2	2	High	3	4	9	5
Overcast	4	0	Mild	4	2	Normal	6	1		
Rainy	3	2	Cool	3	1					
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5		
Rainy	3/9	2/5	Cool	3/9	1/5					

Test instance $\bar{X} = (\text{Outlook}=\text{overcast}, \text{Temp}=\text{cool}, \text{Humidity}=\text{high}, \text{Windy}=\text{true})$

$$P(\text{Play}=\text{no} \mid \bar{X}) \propto P(\bar{X} \mid \text{Play}=\text{no}) P(\text{Play}=\text{no})$$

$$= P(\text{Outlook}=\text{overcast} \mid \text{Play}=\text{no}) \times P(\text{Temp}=\text{cool} \mid \text{Play}=\text{no})$$

$$\times P(\text{Humidity}=\text{high} \mid \text{Play}=\text{no}) \times P(\text{Windy}=\text{true} \mid \text{Play}=\text{no}) \times P(\text{Play}=\text{no})$$

$$= 0 \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0$$

Zero probabilities

- ▶ **Issue:** If the feature value a_i does not co-occur with a class value c , then the corresponding probability estimation will be 0.
- ▶ For example: Given Outlook = overcast, the probability of Play = no is 0/5. The other features will be ignored as the final result will be multiplied by 0.
- ▶ This is bad for our 4 feature dataset, but terrible for (say) a 1000 feature dataset.
- ▶ In text classification, we often encounter situations where a feature does not occur in a particular class.

Laplace smoothing

- ▶ We can “borrow” some probabilities from high probability features and distribute them among zero probability features to avoid having features with zero probabilities.
- ▶ This is called *smoothing*.
- ▶ There are numerous smoothing techniques based on different policies. As long as the total probability mass remains unchanged any policy of probability reassignment is valid.
- ▶ A popular method is called **Laplace smoothing**. For feature x_i and class c the probability estimates will be updated as follows:

$$P(x_i = a \mid C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

where $n(a, c)$ is the number of training objects in class c with $x_i = a$, $N(c)$ is the total number of training objects in class c , and m_i is the number of possible values of feature x_i .

Laplace smoothing: before

$$P(x_i = a | C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

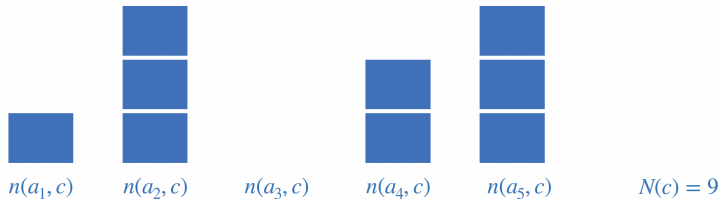


Figure: Laplace smoothing: before

Laplace smoothing: after

$$P(x_i = a \mid C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

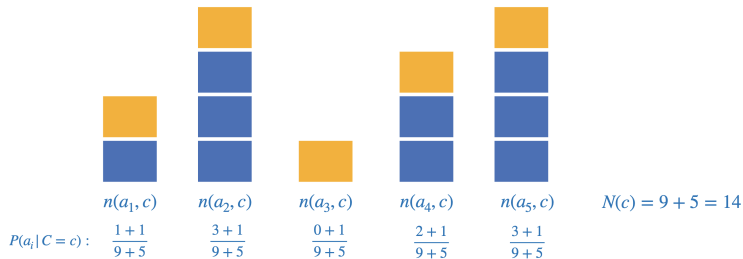


Figure: Laplace smoothing: after